

Highlights

Subseasonal Temperature Forecasting in Andean Stations: A Benchmark of Machine Learning and Deep Learning Models with Anomaly Decomposition

- Zero-leakage climatology isolates thermal anomaly signal
- Sliding 2025 windows benchmark ensembles vs LSTM vs TFT
- Public IGP Huayao hourly data enables full replication
- Ablation quantifies physical feature group contributions
- Probabilistic skill supports subseasonal planning use

Subseasonal Temperature Forecasting in Andean Stations: A Benchmark of Machine Learning and Deep Learning Models with Anomaly Decomposition

Abstract

TODO: max 250 words. Must state (1) AI contribution + (2) engineering application. No undefined acronyms in title/abstract. Stand-alone, no references unless full author+year. Source data: data/raw/dataset.csv (IGP Huayao 2018-2025, gitignored).

Keywords: subseasonal forecasting, anomaly decomposition, machine learning, deep learning

1. Introduction

TODO: objectives only; background adequate, no detailed literature review, no results summary.

2. Methods

TODO: reproducible by an independent researcher. Source: data/raw/dataset.csv (copied from C20-202610-temperature/data/EstacionEMA_2018_2025.csv, 70128 rows + header, Huayao 2018-2025, gitignored per .gitignore:46,52). Summarize published methods + cite; quote with “ ” + citation; describe modifications.

2.1. Data and climatological decomposition

TODO: hourly TT/HR/PP/RR/FF/DD, zero-leakage climatology from train block only.

2.2. Models and validation protocol

TODO: ensembles + LSTM + TFT, sliding multi-window 2025, horizon 720 h.

3. Results

TODO: clear, concise, reproducible. Cite every table/figure. Do not duplicate table data in text.

4. Discussion

TODO: significance, not repetition. Avoid extensive citations here.

5. Conclusions

TODO: main conclusions; may be stand-alone or subsection of Discussion.

6. CRediT authorship contribution statement

TODO: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review and editing. Corresponding: moises.

7. Declaration of competing interests

TODO: “The authors declare no competing interests” or complete declarations tool .docx.

8. Funding

TODO: e.g. “This work was funded by SENATI, Peru (Project C20-2026).” State sponsor role or no involvement.

9. Data statement (Option C, required)

TODO: deposit dataset in repository, cite+link as [dataset] entry in references.bib; or state why not shareable. This statement appears on ScienceDirect.

10. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

TODO: if used: “During preparation ... used [TOOL] for [REASON] ... reviewed/edited ... full responsibility.” If nothing to disclose, omit. Never list AI as author. Disclose AI images per-caption + here.

11. Acknowledgements (TITLE PAGE ONLY - remove from anonymized manuscript)

TODO: keep here for final; exclude from review version.

12. Appendix A

TODO: Eq. (A.1), Table A.1, Fig. A.1 numbering if needed.

References